SHAGGAR INSTITUTE OF TECHNOLOGY
Excellence in Science and Technology

# Advanced Physics (Calculus Based) - Course Syllabus

## 1. Basic Course Information

Course Code and Title:

Advanced Physics 101

Credit Hours:

Instructor(s):

Lectures by Solomon Bililign, Discussion and recitation with TA's TBA

Semester/Quarter:

1st Quarter

Prerequisites:

High school physics/ good mathematical background

Co-requisites:

Class Schedule:

Tuesdays 8-10:30 Eastern US time

Office Hours:

With T/As

Contact Information:

Bililignsol@gmail.com

## 2. Course Description

Provide a brief description of the course, including scope, purpose, and relevance to the program.

Mechanics: Linear Motion in one and two dimensions; Newton's Laws of Motion and their applications to our everyday world; Mechanical Energy--potential and kinetic; Energy transfer; Work. Rotational Motion. The Gravitational Field; the Law of Universal Gravitation; Kepler's Laws of Planetary Motion; the Hydrogen atom. Fluids. Oscillatory Motion: Harmonic Motion; Pendulum and Damped Oscillations. Mechanical Waves: Reflection and Transmission of Waves. Thermodynamics: Temperature and Kinetic Theory of Gases; Thermal Energy and Heat; the Laws of Thermodynamics and their applications

## 3. Course Objectives

List the broad goals of the course and how they align with the program educational objectives.

1. Students will demonstrate knowledge of kinematics by solving problems, class participation and investigation of real-life applications of the physical concepts. They will be able to use vector algebra in solving problems. They will distinguish kinematics in one and two dimensions and rotational kinematics, and apply it to motion of fluids.
2. Students will demonstrate knowledge of laws of motion in describing motion in one dimension and rotational motion and by solving problems, class participation and investigation of real-life applications of the physical concepts. Understand conservation laws: Linear momentum and angular momentum
3. Students will demonstrate knowledge of using the laws of motion to explain behavior of gases, atmospheric phenomena, fluid flow by solving problems, class participation and investigation of real-life applications of the physical concepts.
4. Students will demonstrate knowledge of the different forms of energy and the laws of conservation of energy by solving problems, class participation and investigation of real-life applications of the physical concepts.
5. Students will demonstrate knowledge of periodic motion, elastic energy and different forms waves and their applications for example in seismology acoustics etc. By solving problems class participation and investigation of real-life applications of the physical concepts.
6. Students will demonstrate knowledge of Temperature, Heat as a form of Energy and laws of thermodynamics and understand how they are applied in steam engines and refrigeration by solving problems, class participation and investigation of real-life applications of the physical concepts.

## 4. Student Learning Outcomes (ABET Outcomes 1–7)

Upon successful completion of the course, students will be able to:

1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. Apply engineering design to produce solutions that meet specified needs with consideration for public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. Communicate effectively with a range of audiences.
4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments.
5. Function effectively on a team to provide leadership, create an inclusive environment, establish goals, plan tasks, and meet objectives.
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. Acquire and apply new knowledge as needed, using appropriate learning strategies.

SHAGGAR INSTITUTE OF TECHNOLOGY
Excellence in Science and Technology

## 5. Mapping to Program Outcomes

Insert a table mapping each course learning outcome to the corresponding ABET student outcomes (1–7).

## 6. Required Textbooks and References

List the required and recommended textbooks, manuals, and reference materials.

Physics for Scientists and Engineers with Modern Physics, 10th Edition

Serway and Jewett,

## 7. Course Topics and Schedule

Provide a week-by-week or session-by-session outline of topics, labs, assignments, and assessments.

| THE WEEK OF MM/DD/YY | SUBJECT | UNIT LEARNING OUTCOMES (ULO) | HOMEWORK, EXAM Text book chapters |
|---|---|---|---|
| Week 1 | Introduction to Vectors, Vector algebra and calculus, | Explain units of measurement<br>Explain vector algebra<br>Evaluate and understand applications to different areas of Physics | Physics for Scientists and Engineers with Modern Physics, 10th Edition<br>Chapter 3 plus notes<br><br>**Assignment 1-Group project** |
| Week 2 | Vector calculus, vector differentiation , line, surface and volume integrals, applications to Physics | Definition and explain acceleration<br>Evaluate kinematics equations in one-dimensional horizontal<br>Evaluate Kinematics in two-dimension under the action of gravity | Physics for Scientists and Engineers with Modern Physics, 10th Edition<br>Chapter 3 plus notes<br><br>**Assignment 2-Group Project**<br>**Quiz 1** |

| | | | |
|---|---|---|---|
| Week 3 | Kinematics: Motion in one and two dimensions<br>Circular motion | Evaluate two-dimensional kinematics<br>Explain projectile motion<br>Explain circular motion<br>Explain relative motion | Physics for Scientists and Engineers with Modern Physics, 10th Edition<br>Chapter 2, 3, 6<br><br>**Assignment 3- Group Project**<br>**Quiz 2** |
| | | | Exam #1 (First 3 weeks) |
| Week 4 | Dynamics<br>Force and Newton's Laws of Motion.<br>Dynamics of linear motion, Gravitation, statics | Explain concepts of inertia and mass<br>Explain concepts of force and net unbalanced force<br>Explain Newton's first law of motion | Physics for Scientists and Engineers with Modern Physics, 10th Edition<br>Chapter 5,6,12 and 13<br><br>**Assignment 4- Group Project**<br>**Quiz 3** |
| Week 5- | Dynamics<br>Circular Motion and Other Applications of Newton's Laws<br>Rotational Motion | Explain the concept of rotational Inertia<br>Explain the concept of Torque<br>Evaluate problems on Newton's laws in rotational motion<br>Describe uniform circular motion<br>Define and Describe pressure and density<br>Explain hydrostatics and Archimedes principle<br>Explain Pascale's principle<br>Describe fluid dynamics and equation of continuity<br>Describe fluid dynamics-Bernoulli's equation | Physics for Scientists and Engineers with Modern Physics, 10th Edition<br>Chapter 5,6,10, 12,13<br><br>**Assignment 5- Group Project**<br>**Quiz 4** |
| | | | |

| | | | |
|---|---|---|---|
| Week 6 | Work and Energy Conservation of Energy, Rotational Energy, and Power | Explain work by a constant force<br>Explain concept of energy and forms of energy: kinetic and potential<br>Evaluate work energy theorem<br>Describe conservation of mechanical energy<br>Evaluate problems in conservation of energy Evaluate work by a variable force<br>Evaluate rotational energy (rotational kinetic energy)<br>Explain work by a constant force | Physics for Scientists and Engineers with Modern Physics, 10th Edition<br>Chapter 7,8<br><br>**Assignment 6- Group Project**<br>**Complete:** Test #1 |
| | | | |
| Week 7 | Impulse, Linear, and Angular Momentum | Define and explain impulse and momentum<br>Describe types of collision: elastic and inelastic<br>Describe conservation of momentum<br>Explain angular momentum<br>Evaluate problems on conservation of angular momentum | Physics for Scientists and Engineers with Modern Physics, 10th Edition<br>Chapter 9,11<br><br>**Assignment 7- Group Project**<br>**Quiz 5** |
| Week 8 | Unit 11: Simple Harmonic Motion and Elasticity. | Explain simple harmonic motion<br>Explain the concepts of frequency and period<br>Describe elastic potential energy<br>Describe damped motions | Physics for Scientists and Engineers with Modern Physics, 10th Edition<br>Chapter 15,16<br><br>**Assignment 8- Group Project**<br>**Quiz 6** |

| | | | |
|---|---|---|---|
| | | Explain the relation to circular motion | |
| | | Exam 2 (week 5 to 8 material) | |
| Week 9 | Waves and Sound | Explain the concept of waves<br>Describe transverse and longitudinal waves<br>Describe wave concepts: amplitude and frequency<br>Explain sound waves and sound intensity<br>Explain Doppler effect and applications | Physics for Scientists and Engineers with Modern Physics, 9th Edition<br>Chapter 17<br><br>**Assignment 9- Group Project**<br>**Exam 2** |
| Week 10 | Temperature, Expansion and Calorimetry | Describe temperature, and measurement I scales<br>Explain the constant volume gas thermometer and the kelvin scale Describe specific heat capacity<br>Explain Calorimetry<br>Describe fusion, vaporization, and latent heat of fusion | Physics for Scientists and Engineers with Modern Physics, 9th Edition<br>Chapter 18,19<br><br>**Assignment-10- Group Project**<br>**Quiz 7** |
| Week 11 | Laws of Thermodynamics<br>Kinetic theory | Describe Ideal gas laws<br>Explain heat, work, and internal energy of a gas<br>Describe the laws of thermodynamics: 0thLaw and First law<br>Describe and evaluate Thermodynamic processes: isobaric, isochoric, isothermal, and adiabatic<br>Describe Ideal gas laws, | Physics for Scientists and Engineers with Modern Physics, 10th Edition<br>Chapter 20,21<br><br>**Assignment 11- Group Project**<br>**Quiz 8** |

| | | | |
|---|---|---|---|
| Week 12- | | Review | |
| | | | |

## 8. Laboratory Component
Describe lab activities, equipment used, safety requirements, and expected outcomes.

## 9. Grading Policy

Breakdown of grades: assignments, quizzes, lab work, midterms, final exams, participation, etc.

| Category | # of Activities | Percentage Grade Weight |
|---|---|---|
| Assignment | 11 | 33% |
| Quizzes | 8 | 24% |
| Test | 2 | 20% |
| Final Exam | 1 | 23% |
| **Total** | | **100%** |

## 10. Academic Integrity
State the academic honesty policy and consequences in accordance with institutional regulations.

## 11. Accessibility and Student Support
Information on disability services, tutoring, counseling, and academic support resources.

## 12. Safety Considerations
Outline safety guidelines for labs, fieldwork, or other activities.